WEEK 8

Data Science I

Dataset, exploratory data analysis (EDA) and feature engineering

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Goal of this week

1

Data

UCI Student Performance downloaded; sha256 checksums stored in CHECKSUMS.txt and verified on every run.

2

EDA

6 figures, eda_summary.json and two notebooks (TR/EN). Every number is computed in code.

3

Features

EARLY and MIDTERM feature sets; an unfitted preprocessing pipeline.

> *No model was trained this week. Model comparison, cross-validation and SHAP come in Week 9.*

Dataset

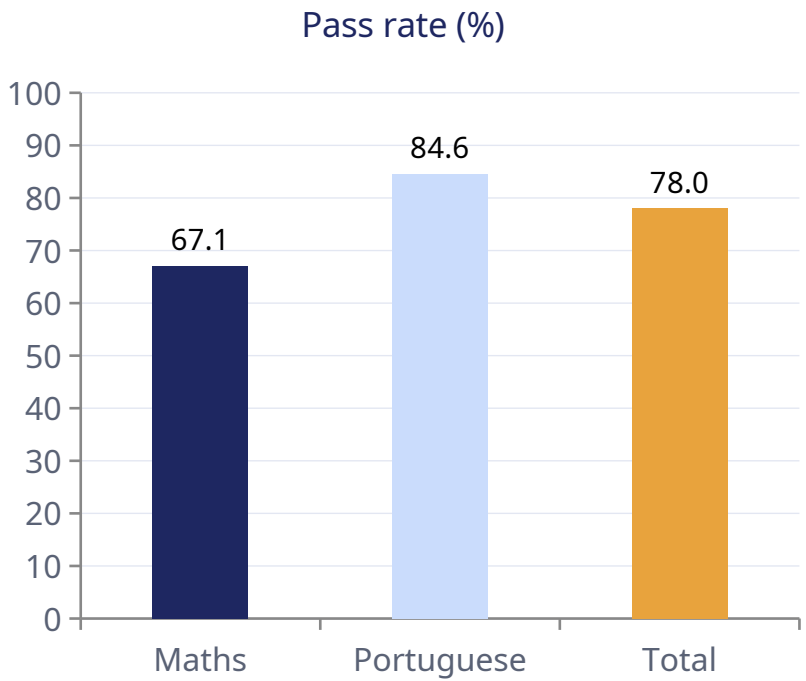
1044	395 / 649	0	366
total records	Maths / Portuguese	missing values	students in both subjects

Source: Cortez & Silva (2008), UCI Machine Learning Repository, CC BY 4.0. Two Portuguese secondary schools; grade scale 0–20 (G1, G2 in-term; G3 final). Shared students are marked with student_key.

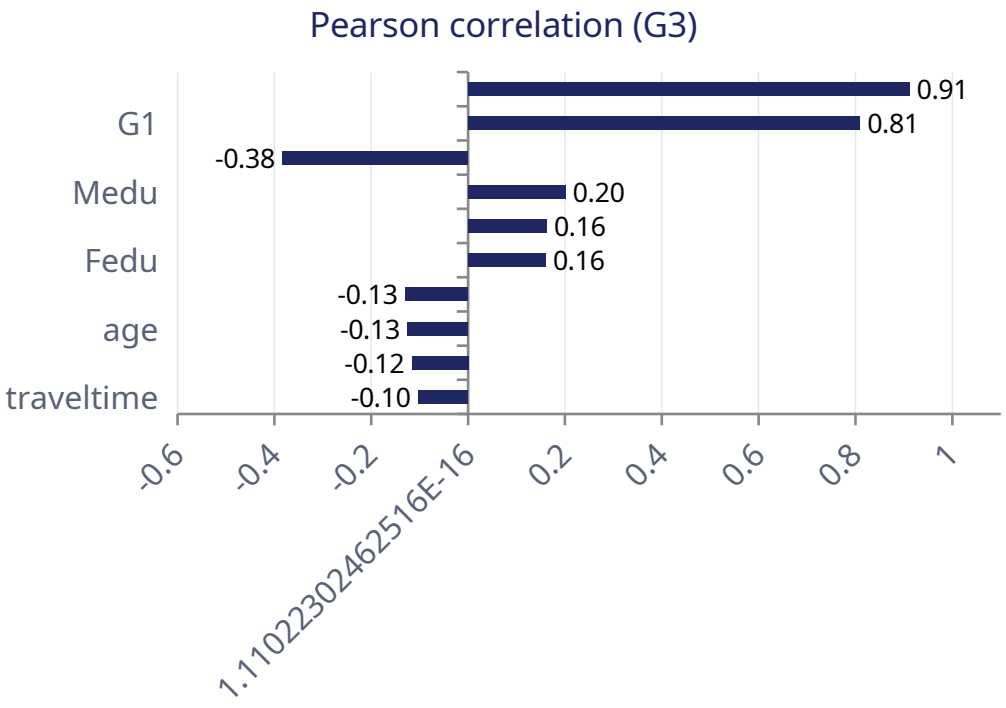
Target definition: at_risk = G3 < 10

```
G3 < 10   ⇔   G3 × 5 < 50 = PASS_GRADE
```

The threshold matches our system's pass grade (PASS_GRADE=50) exactly. Risk rate is 32.9% in Maths and 15.4% in Portuguese; 230 students overall (22.0%).



Top 10 relationships with G3



- Strongest non-grade signal: past failures (failures, −0.38).
- Parental education and study time: weak positive (0.16–0.20).
- Alcohol, age and travel time: weak negative (−0.10 to −0.13).

Leakage risk: G1 and G2

0.91

G2 – G3

0.81

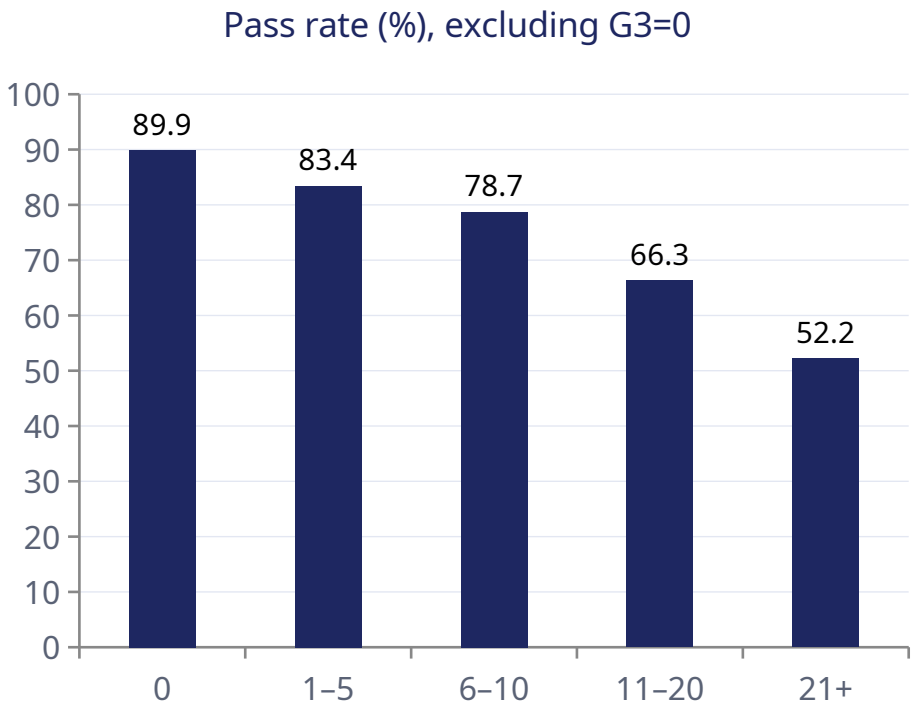
G1 – G3

Predicting G3 from G2 is almost repeating the outcome. Two separate feature sets were therefore defined:

EARLY	No G1 or G2 — start-of-term early warning
MIDTERM	G1 included, no G2 — mid-term

G3 is in neither set; this is enforced by tests.

Absence: corrected finding



- All 53 G3=0 records have 0 absences; they suppressed the first correlation (Pearson −0.05).
- Excluding G3=0: Pearson −0.22, Spearman −0.24. Pass rate falls steadily as absence rises.
- Still, half of the 21+ group passes (n=23): absence alone does not justify a decision.
- Our system should use unexcused absences.

Bug record 1: absence analysis

Field	Description
Symptom	In the first EDA, absence was not among the 10 strongest relationships with G3; it was concluded to be a "weak indicator".
Root cause	All 53 G3=0 records fall in the 0-absence bucket. These records (possible missed exam/dropout) suppressed the relationship.
Fix	Correlation and bucket analysis computed separately with and without G3=0 (notebook, section f).
Status	Verified: excluding G3=0, Pearson −0.22; pass rate falls from 89.9% to 52.2% across buckets.

Data quality and caveats

Topic	Value	Decision
G3 = 0	`53 records (5.1%)`	Not removed; a g3_zero flag was added. All have 0 absences; the reason (missed exam, dropout) is not stated. Relationship analyses should also be run without them.
Shared students	`366`	The literature reports 382; matching columns differ. Used as GroupKFold groups in Week 9.
Missing values	`0`	No imputation needed.

Feature engineering

Derived features

- parent_edu_max — higher of parents' education
- alcohol_total — Dalc + Walc
- support_any — school or family support
- has_failures — any past failure
- absences_log1p, absence_bucket

Preprocessing pipeline

- Binary categories as 0/1
- Nominal categories one-hot (handle_unknown="ignore")
- Ordinal columns kept as is
- Returned unfitted; fitted inside CV folds in Week 9 (no leakage)

Mapping to our system

UCI feature	Our schema	Status
`G1 / G2`	Grade average ÷ 5	**Derivable**
`absences`	Attendance records (with excuse flag)	**Available**
`failures`	—	**Missing**
`studytime, Medu/Fedu, support`	—	**Missing**

Takeaway: our system only stores grades and attendance. The Week 10 risk indicator will therefore be an attention flag shown with its reasons, not a verdict.

Verification

233 passed

1 warning (not yet reviewed)

```
python -m data_science.download
python -m data_science.eda
python -m data_science.features
```

- Tests use no network; they run on a 30-row synthetic fixture CSV.
- Both notebooks executed from scratch without errors via nbconvert.
- Checks: target threshold (G3=9 → 1, G3=10 → 0), no G1/G2/G3 in the sets, NaN-free transform, unknown category.
- Application code, mock data and eval were kept out of scope this week.

Notebooks

`01_uci_eda.ipynb`	Turkish narrative
`01_uci_eda_eng.ipynb`	English narrative
`notebooks/html/`	HTML for viewing outside GitHub

Notebooks were executed from scratch and saved with outputs. Logic is imported from the data_science package; no code was copied. Numbers are not typed into markdown; they come from cell outputs.

a. Dataset
b. Target definition
c. G3 distribution
d. Correlations
e. Leakage
f. Absence analysis
g. Feature sets
h. Conclusion

UCI's limits and the decision

What UCI provides: a general risk signal

- Early grades strongly reflect the final (G1–G3 r = 0.81)
- Past failures: strongest non-grade signal (r = −0.38)
- Absence: meaningful excluding G3=0, but not decisive alone

What UCI lacks: context

- One school year, two subjects, two schools
- No course, teacher or exam context
- No exam dates or order; no real early warning
- ~678 unique students, ~230 at risk: small sample

Decision: UCI stays as the general (baseline) model; the context hypothesis is tested on a larger, time-stamped dataset.

Hypothesis: risk depends on course context

> *Example: one teacher's final exam was harder than others last year and many students' grades dropped. A student who was borderline in the first two exams is at higher risk in that course than a student with the same grades in another course.*

Week 8
UCI: general model baseline

›

Week 9
OULAD: course-history features

›

Week 10
Compare: general vs course-history

›

Week 11
Recipe into the system: risk tab

Principles: risk is an attention flag with reasons, not a verdict. No sociocultural features in the system; only grades, unexcused absence and course context. No data or model moves from OULAD into the system, only the validated feature recipe.

Open items and Week 9

Open items

- Bug record 1 (absence analysis): verified.
- Alternative text missing on notebook figures (6 images).
- Week 6 bug 4 (silent averaging in pivot): to be verified.
- Week 7 eval: m5 (causal speculation) and c4 (73.98 → "73") still open.
- The pytest warning will be reviewed.

Week 9 — OULAD

- Larger, time-stamped data: OULAD (Open University)
- Course-history features: a grade relative to the course's past average
- Temporal split: train on 2013 presentations, test on 2014
- Presented in notebook 02